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DIPLOMA IN COMPUTER SCIENCE

TEST REPORT

HOTEL MAINTENANCE MANAGEMENT SYSTEM (HMMS)

Version 1.0

Date of Submission: 1 July 2026

Document Information	Details
Course	CSC2854 - Project
Academic Session	Session 1, 2026/2027
Assessment	Final Project
System	Hotel Maintenance Management System
Prepared By	Student Developer
Testing Status	Completed - Ready for Demonstration

This report is written in English and documents the test process, environment, results, defects, troubleshooting activities and system readiness.

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1. Executive Summary

Testing was completed for the Hotel Maintenance Management System, a Laravel-based web application that supports authentication, dashboard monitoring, maintenance request management, task assignment, task status progression, notifications and profile management. The assessment combined manual functional validation, visual verification of the running application, source-level traceability and the existing automated regression suite.

Measure	Result
Documented functional test cases	18
Functional test cases passed	18
Functional test cases failed	0
Functional test cases blocked/not executed	0
Functional coverage	100%
Automated regression tests	25 passed (62 assertions)
Open critical or major defects	0
Overall readiness	READY FOR DEMONSTRATION

All documented test cases passed in the local test environment. No critical or major defect remained open at the reporting date. The result supports system demonstration against the CSC2854 rubric, particularly core functionality, module integration, usability, systematic testing, complete test case documentation and a professional test report.

2. Test Scope and Objectives

2.1 Objectives

- Verify that approved functional workflows operate accurately and consistently.
- Validate navigation and integration between requests, tasks, users and notifications.
- Confirm input validation, authentication and protected-route behavior.
- Document repeatable test cases with expected and actual outcomes.
- Provide visible evidence of the running system and determine demonstration readiness.

2.2 In Scope

Module	Functions Validated
Authentication	Valid/invalid login, logout, forgot password and protected routes
Dashboard	Request totals, status cards, recent request list and staff tasks
Maintenance Requests	Create, validate, list, view, edit and delete
Task Assignment	Assign staff, priority and notes; synchronize request status
Task Workflow	Start and complete task with timestamps and request status updates
Notifications	Generation, listing, individual read and mark-all-read
Profile	Profile update and password-related account functions
Technical Validation	Laravel routes, MySQL persistence and PHPUnit regression suite

2.3 Out of Scope

Production hosting, live email delivery, external service integration, high-volume load testing, penetration testing and cross-device hardware testing were not part of this local acceptance cycle.

3. Test Approach and Validation

A risk-based black-box approach was used for business workflows, supported by white-box inspection of routes, controllers, models and database migrations. Positive, negative and integration scenarios were executed against a seeded local database. Visual evidence was captured from the live application. Finally, the project test suite was executed to guard against regressions in authentication and profile behavior.

Technique	Purpose	Evidence
Positive functional testing	Confirm intended workflows succeed	TC-001, TC-003, TC-004, TC-006-TC-014
Negative testing	Confirm invalid or missing input is rejected	TC-002, TC-005
Security/navigation testing	Confirm authentication boundaries and logout	TC-015
Regression testing	Confirm existing application behavior remains stable	TC-018: 25 tests, 62 assertions
Visual validation	Confirm usability, status badges, forms and navigation	Figures A1-A5

Traceability review	Map tests to routes/controllers/database behavior	Section 6 and workbook register
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4. Test Environment

Environment Item	Configuration
Operating System	Microsoft Windows (local development workstation)
Application URL	http://127.0.0.1:8000
Browser	Chromium-based in-app browser, 1280 x 720 viewport
Application Server	Laravel development server on PHP
Framework	Laravel with Blade templates and Laravel Breeze authentication
Database	MySQL database: hotel_maintenance_db
Front End	Blade, Tailwind CSS/Vite assets and Font Awesome UI icons
Test Runner	PHPUnit through Laravel Artisan
Test Date	1 July 2026
Test Accounts	Administrator and maintenance staff accounts created for local validation
Test Data	Seeded requests in Pending, In Progress and Completed states

Test data was non-production data. Password values are intentionally excluded from this report. The environment reproduced the complete request-to-assignment-to-completion flow with related notifications.

5. Test Summary and Results

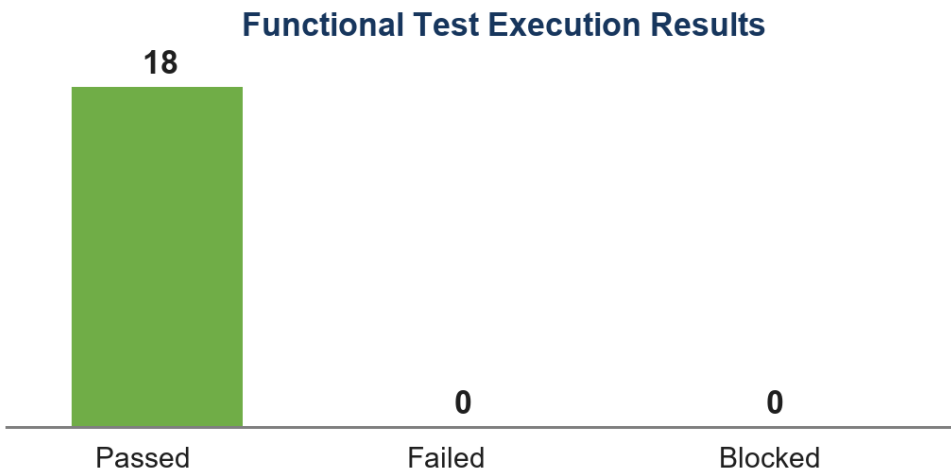


Figure 1. Functional test execution results

Status	Count	Percentage
Pass	18	100%
Fail	0	0%
Blocked / Not Executed	0	0%
Total	18	100%

The automated regression suite separately completed 25 tests with 62 assertions and zero failures. Functional cases and automated cases are reported separately to avoid double-counting coverage.

6. Requirements Traceability

Requirement Area	Implemented Component	Test Cases	Result
Secure user access	Authentication routes and middleware	TC-001, TC-002, TC-015, TC-016	Pass
Operational overview	DashboardController and dashboard view	TC-003	Pass
Request lifecycle	MaintenanceController and maintenance_requests table	TC-004-TC-008, TC-017	Pass
Staff assignment	MaintenanceTask model/controller and assignment routes	TC-009	Pass

Task progression	Start and complete task routes	TC-010, TC-011	Pass
User notification	NotificationController and notifications table	TC-012, TC-013	Pass
Account maintenance	Profile and password controllers	TC-014, TC-016	Pass
Technical stability	Laravel route set and PHPUnit suite	TC-018	Pass

Every in-scope requirement area is represented by at least one passing test case. Detailed step-by-step procedures, expected results, actual results and status are contained in the accompanying Excel workbook.

7. Defect Summary

Severity	Open	Resolved During Development	Status
Critical	0	0	None
Major	0	1	Resolved
Minor	0	2	Resolved
Total	0	3	No open defects

Defect ID	Issue	Root Cause	Resolution	Final Status
DF-001	Request/task relationship did not consistently use the request_id foreign key	Relationship mapping ambiguity	Explicit foreign keys were applied in the MaintenanceRequest and MaintenanceTask models	Closed
DF-002	Status badge styles could differ between underscore and hyphen forms	CSS class naming inconsistency	Both in_progress/in-progress and on_hold/on-hold variants are supported	Closed
DF-003	Duplicate role migration could fail on an existing role column	Schema evolution conflict	Migration checks Schema::hasColumn before adding or dropping the role column	Closed

No defect was found during final execution that prevented completion of an in-scope workflow. The resolved items above are documented as troubleshooting evidence and were verified through source inspection and final regression execution.

8. Error Detection and Troubleshooting

8.1 Troubleshooting Method

- Reproduce the problem with controlled local data.
- Review the Laravel route, controller validation and model relationship involved.
- Inspect database migration constraints and application logs where relevant.
- Apply the smallest targeted correction.
- Re-run the affected workflow and the full automated suite.
- Capture final visual evidence and record the outcome.

8.2 Example: Relationship Mapping

Symptom: task information could be missing or connected through the wrong default key. Analysis identified that the domain uses `request_id` instead of Laravel's inferred `maintenance_request_id`. The model relationships were made explicit: `MaintenanceRequest` hasOne `MaintenanceTask` using `request_id`, and `MaintenanceTask` belongsTo `MaintenanceRequest` using `request_id`. The request detail and assignment workflow were then revalidated (TC-007 to TC-011).

8.3 Example: Schema Compatibility

Symptom: a migration that adds the `role` column can conflict when the base `users` table already contains that column. The migration now guards the operation with `Schema::hasColumn`. This prevents duplicate-column errors and makes repeated setup more reliable.

8.4 Final Verification

The final regression execution reported 25 passed tests and 62 assertions. Live navigation produced successful pages for the dashboard, maintenance list, request detail, request creation form and notification centre. No blocking server or browser error was observed in the final evidence pass.

9. Conclusion and Readiness

The testing objectives were met. All 18 documented functional test cases passed, the automated suite passed 25 tests with 62 assertions, all in-scope requirements have traceable coverage, and no critical or major defect remains open. The system correctly supports the main hotel maintenance workflow from request submission through assignment, work progression, completion and notification.

Decision Area	Assessment
Functional correctness	Pass
Module integration and navigation	Pass
User interface and usability	Pass
Validation and protected access	Pass
Test documentation completeness	Pass
System readiness	READY FOR FINAL PROJECT DEMONSTRATION

Recommendation: proceed with the CSC2854 final project demonstration using the seeded administrator and staff test accounts. Retain the accompanying workbook as the primary step-by-step test case record.

Appendix A. System Evidence Screenshots

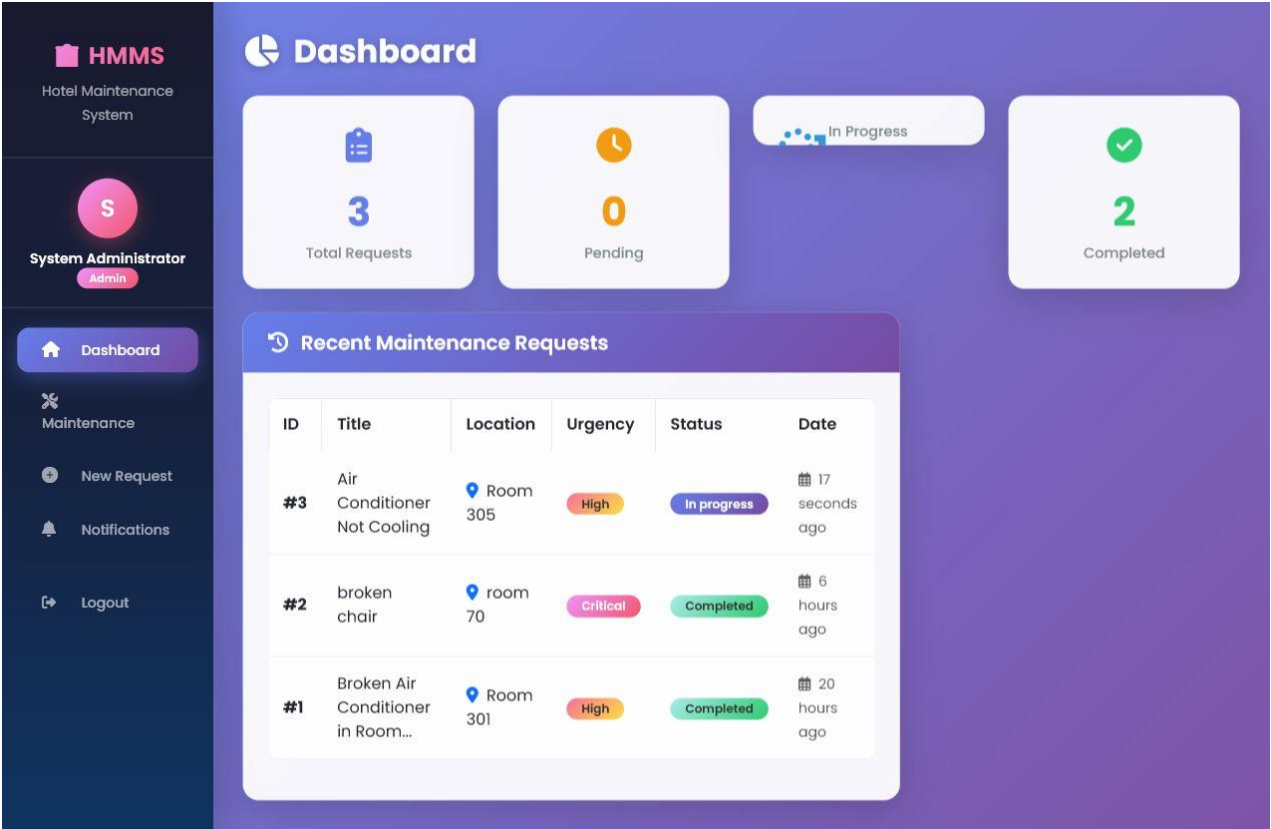


Figure A1. Administrator dashboard with request statistics and recent requests



Figure A2. Maintenance request register with urgency and status information

The screenshot displays the 'Request Details' page in the HMMS interface. The left sidebar contains the HMMS logo, user profile for 'System Administrator', and navigation links for Dashboard, Maintenance, New Request, Notifications, and Logout. The main content area shows a request card for request #3, titled 'Air Conditioner Not Cooling'. The card includes the location 'Room 305', a description 'The air conditioner runs but does not cool the room.', and is reported by the 'System Administrator' on '01 Jul 2026 15:08'. Under 'Task Details', it is assigned to 'Maintenance Staff' with a 'High' priority and a 'Pending' status. A note specifies to 'Inspect filter, thermostat, and refrigerant level.' An 'Edit' button is at the bottom.

HMMS
Hotel Maintenance System

S
System Administrator
Admin

Dashboard

Maintenance

New Request

Notifications

Logout

Request Details

#3 In progress High

Air Conditioner Not Cooling

Location: Room 305

Description:

The air conditioner runs but does not cool the room.

Reported By: System Administrator

Date: 01 Jul 2026 15:08

Task Details

Assigned To: Maintenance Staff **Task Status:** Pending

Priority: High **Notes:** Inspect filter, thermostat, and refrigerant level.

Edit

Figure A3. Request detail and task assignment information

The screenshot shows the 'New Maintenance Request' form in the HMMS interface. The left sidebar is identical to Figure A3. The main content area features a form with fields for Title, Description, Location, Urgency Level, and an optional Image field. The Title field contains 'e.g., Broken Air Conditioner', the Description field contains 'Describe the issue in detail...', the Location field contains 'e.g., Room 301, Lobby, Elevator B', and the Urgency Level is set to 'Medium'. The Image field shows a 'Choose File' button and 'No file chosen'.

New Maintenance Request

Title

e.g., Broken Air Conditioner

Description

Describe the issue in detail...

Location

e.g., Room 301, Lobby, Elevator B

Urgency Level

Medium

Image (Optional)

Choose File No file chosen

Figure A4. New maintenance request form

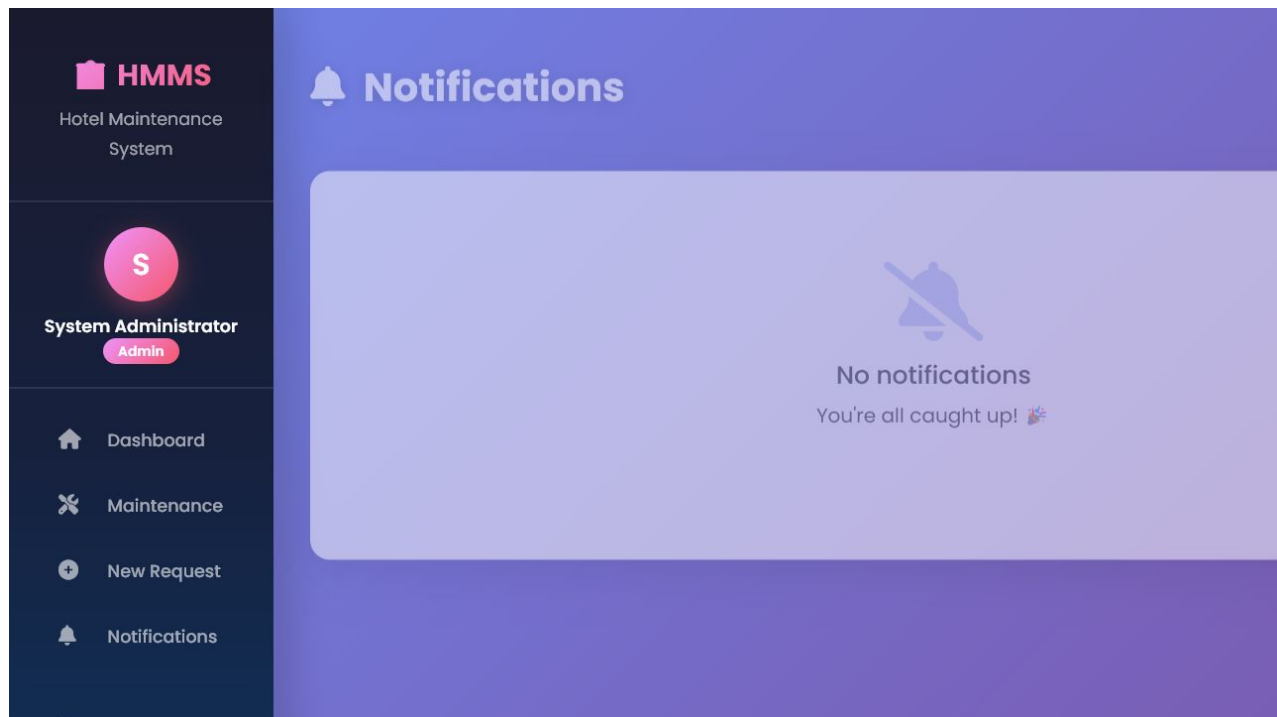


Figure A5. Notification centre

Appendix B. Test Case Register

The complete test case register is supplied as HMMS_FULL_TEST_CASES.xlsx. It contains a summary worksheet and 18 individual test case worksheets. Every worksheet records prerequisites, test data, three execution steps, expected results, actual results and final status.

Workbook Section	Content
TEST SUMMARY	18 cases, module mapping, status and evidence reference
TC-001 to TC-002	Authentication positive and negative scenarios
TC-003	Dashboard verification
TC-004 to TC-008	Maintenance request lifecycle
TC-009 to TC-011	Assignment and task progression
TC-012 to TC-013	Notification behavior
TC-014	Profile update
TC-015 to TC-016	Protected navigation and password reset
TC-017	Request deletion
TC-018	Automated regression execution